

Data Mining Assignment

Groceries: Association Rule Analysis

Name: SHEMA AIMABLE

Student number: 101503

Method

The analysis uses the groceries.csv dataset and the Apriori algorithm from the R arules package [1]. Each row represents one shopping basket. Before the analysis, blank cells and extra spaces in item names were removed. Rows with the same items were kept because they can represent separate purchases.

The minimum support was set to 1% and the minimum confidence to 30%. This keeps combinations found in at least 1% of baskets, and rules where at least 30% of baskets with the antecedent also contain the consequent. The assignment did not give specific thresholds.

Question 1: Understand the Data

- The dataset contains 9,835 transactions. Each transaction represents one shopping basket.
- The dataset contains 169 different grocery items.
- The ten most frequently purchased items are listed in Table 1, from the highest to the lowest basket count.

Table 1: Most frequently purchased items

Item	Baskets	Percentage of baskets
whole milk	2,513	25.55%
other vegetables	1,903	19.35%
rolls/buns	1,809	18.39%
soda	1,715	17.44%
yogurt	1,372	13.95%
bottled water	1,087	11.05%
root vegetables	1,072	10.90%
tropical fruit	1,032	10.49%
shopping bags	969	9.85%
sausage	924	9.40%

Whole milk is the most frequently purchased item. It appears in 2,513 baskets, which is 25.55% of the dataset, or about one in four baskets.

Question 2: Frequent Itemsets

- The analysis found 245 frequent itemsets containing at least two products. Ten of these itemsets are listed in Table 2. A + sign joins products found in the same basket.
- The support of each itemset is shown in the last column of Table 2. Support is calculated by dividing its basket count by 9,835 and multiplying by 100.

Table 2: Frequent itemsets

Itemset	Baskets	Support
other vegetables + whole milk	736	7.48%
rolls/buns + whole milk	557	5.66%
whole milk + yogurt	551	5.60%
root vegetables + whole milk	481	4.89%
other vegetables + root vegetables	466	4.74%
other vegetables + yogurt	427	4.34%
other vegetables + rolls/buns	419	4.26%
tropical fruit + whole milk	416	4.23%
soda + whole milk	394	4.01%
rolls/buns + soda	377	3.83%

c. Other vegetables + whole milk has the highest support among combinations with two or more products. The two items appear together in 736 baskets, so their support is $736 / 9,835 = 7.48\%$.

d. A support of 7.48% means that about 7 or 8 baskets out of every 100 contain both other vegetables and whole milk. Support shows how common a product combination is in the dataset.

Question 3: Association Rules

a. The analysis generated 125 association rules. Ten are listed in Table 3: the eight with the highest support and the two with the highest confidence and lift.

b. Table 3 reports the antecedent, consequent, support, confidence and lift for each rule. The antecedent is the condition of the rule, and the consequent is the item associated with that condition.

Table 3: Association rules

Rule	Antecedent	Consequent	Support	Confidence	Lift
1	other vegetables	whole milk	7.48%	38.68%	1.514
2	rolls/buns	whole milk	5.66%	30.79%	1.205
3	yogurt	whole milk	5.60%	40.16%	1.572
4	root vegetables	whole milk	4.89%	44.87%	1.756
5	root vegetables	other vegetables	4.74%	43.47%	2.247
6	yogurt	other vegetables	4.34%	31.12%	1.608
7	tropical fruit	whole milk	4.23%	40.31%	1.578
8	tropical fruit	other vegetables	3.59%	34.21%	1.768
9	citrus fruit + root vegetables	other vegetables	1.04%	58.62%	3.030
10	citrus fruit + other vegetables	root vegetables	1.04%	35.92%	3.295

Question 4: Understanding the Measures

Support is the percentage of all baskets containing the items on both sides of a rule.

Confidence is the percentage of antecedent baskets that also contain the consequent. Lift compares this confidence with the consequent's percentage in the whole dataset [2, 3]. The following three rules illustrate the difference.

Rule 1: other vegetables -> whole milk

- a. Support is 7.48%. Both products occur in 736 of the 9,835 baskets, so support is $736 / 9,835 = 7.48\%$.
- b. Confidence is 38.68%. Of the 1,903 baskets containing other vegetables, 736 also contain whole milk. Confidence is $736 / 1,903 = 38.68\%$.
- c. Lift is 1.514. The overall percentage for whole milk is 25.55%. Dividing confidence by this percentage gives the lift, using unrounded values. The consequent is about 1.51 times as common in baskets containing the antecedent as in the whole dataset.

Rule 3: yogurt -> whole milk

- a. Support is 5.60%. Both products occur in 551 of the 9,835 baskets, so support is $551 / 9,835 = 5.60\%$.
- b. Confidence is 40.16%. Of the 1,372 baskets containing yogurt, 551 also contain whole milk. Confidence is $551 / 1,372 = 40.16\%$.
- c. Lift is 1.572. The overall percentage for whole milk is 25.55%. Dividing confidence by this percentage gives the lift, using unrounded values. The consequent is about 1.57 times as common in baskets containing the antecedent as in the whole dataset.

Rule 5: root vegetables -> other vegetables

- a. Support is 4.74%. Both products occur in 466 of the 9,835 baskets, so support is $466 / 9,835 = 4.74\%$.
- b. Confidence is 43.47%. Of the 1,072 baskets containing root vegetables, 466 also contain other vegetables. Confidence is $466 / 1,072 = 43.47\%$.
- c. Lift is 2.247. The overall percentage for other vegetables is 19.35%. Dividing confidence by this percentage gives the lift, using unrounded values. The consequent is about 2.25 times as common in baskets containing the antecedent as in the whole dataset.

Question 5: Find Interesting Rules

- a. Rule 9, citrus fruit + root vegetables -> other vegetables, has the highest confidence at 58.62%.
- b. Rule 10, citrus fruit + other vegetables -> root vegetables, has the highest lift at 3.295.
- c. Rule 1, other vegetables -> whole milk, has the highest support at 7.48%.

Table 4 summarises these results. They were selected by comparing all 125 generated rules.

Table 4: Rules with the highest measures

Measure	Rule	Association	Highest value
Confidence	9	citrus fruit + root vegetables -> other vegetables	58.62%

Measure	Rule	Association	Highest value
Lift	10	citrus fruit + other vegetables -> root vegetables	3.295
Support	1	other vegetables -> whole milk	7.48%

d. No, these are three different rules. Rule 1 is the most common combination. Rule 9 has the highest percentage of antecedent baskets containing the consequent. Rule 10 has the largest increase compared with the consequent's overall frequency. The measures select different rules because they measure different things.

Question 6: Interpret a Rule

Rule 5, root vegetables -> other vegetables, means that baskets containing root vegetables often also contain other vegetables. Of the 1,072 baskets containing root vegetables, 466 also contain other vegetables. Confidence is therefore 43.47%.

The rule shows a fairly strong positive association. Its lift is 2.247, so other vegetables are about 2.25 times as common in baskets containing root vegetables as in the whole dataset. Both categories appear together in 4.74% of all baskets. However, the rule does not mean that every purchase of root vegetables includes other vegetables.

Question 7: Business Application

- Root vegetables and other vegetables could be placed close together in the produce area. They appear in the same basket 466 times, and their lift of 2.247 shows a positive association.
- A possible bundle would contain root vegetables and other vegetables. For example, the manager could select products from both categories for a mixed vegetable offer and check whether customers buy it.
- When an online basket contains root vegetables but no other vegetables, the supermarket could recommend a product from the other-vegetables category. This follows rule 5, since 43.47% of baskets containing root vegetables also include other vegetables.

Question 8: Understanding Association

- No. An association rule does not prove that buying one product causes someone to buy another. It only shows that the products occur together. Customers may buy both for the same meal, because of a promotion, or because they like both products.
- A rule can have high confidence but low lift when the consequent is already very common. For example, if confidence is 80% but the consequent occurs in 90% of all baskets, lift is $0.80 / 0.90 = 0.889$. The high confidence does not show an increase over the usual purchase rate.
- Lift greater than 1 means the products occur together more often than expected if their purchases were independent. It indicates a positive association, but it does not prove causation.

References

- [1] R arules documentation: Mining Associations with the Apriori Algorithm. <https://search.r-project.org/CRAN/refmans/arules/html/apriori.html>
- [2] IBM Db2 documentation: Characteristics of association rules and item sets. <https://www.ibm.com/docs/en/db2/11.1.0?topic=visualizer-characteristics-association-rules-item-sets>

[3] IBM Db2 documentation: Lift in an association rule.
https://www.ibm.com/docs/en/db2/11.1.0?topic=SSEPGG_11.1.0%2Fcom.ibm.im.model.doc%2Fc_lift_in_an_association_rule.htm